

ICML 2024 | Certify Graph Neural Networks Together, Not One Node at a Time

The first collective robustness certificate against graph injection attacks lifts the certified ratio from 0% to over 80%

Graph neural networks (GNNs) sit behind citation search, recommendation, and fraud detection, yet they are unsettlingly easy to fool. A graph injection attack (GIA) slips a handful of malicious nodes into the graph, wires them to a few real target nodes, and quietly flips the model's predictions.

When a model guards something that matters, we want more than "it seems robust" — we want certified robustness, a mathematical guarantee that the prediction cannot change unless the attacker exceeds a stated budget. The catch is that every existing certificate for injection attacks reasons about one target node at a time. This sample-wise view is far too pessimistic: once the attacker's budget grows even modestly, it certifies almost nothing, and the certified ratio falls to 0.0%.

This ICML 2024 paper from The Hong Kong Polytechnic University starts from a simple observation about how real attacks work. An adversary cannot conjure a fresh graph for every node they want to fool; they inject one perturbed graph, and that single graph has to disrupt the entire set of target nodes at once. Certify the whole target set jointly rather than node by node, and the guarantee should be far stronger. The result is the first collective certified robustness scheme for GNNs against graph injection attacks.

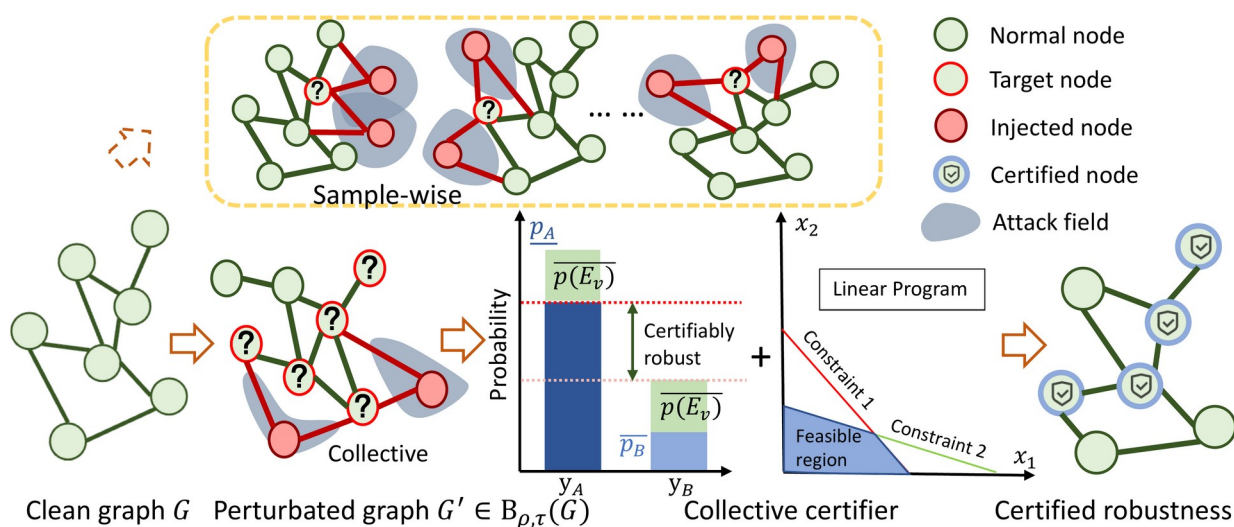


Figure 1. The collective certification idea. Sample-wise certification (top, yellow box) assumes the attacker can craft a separate perturbed graph for each target node, which makes it extremely conservative. The collective scheme (bottom) faces a single perturbed graph and uses one linear program to bound the worst case over all target nodes at once, certifying many more of them (shield markers, right).

Paper: <https://arxiv.org/abs/2403.01423>
<https://github.com/Yuni-Lai/CollectiveLPCert>

Code:

Why certifying one node at a time falls apart

Sample-wise certification bakes in an unrealistic assumption: that the attacker can tailor a separate perturbed graph for each target node. In reality the attacker submits a single graph. The one prior collective scheme (Schuchardt et al., 2020) was designed for edge-modification attacks and assumes a fixed receptive field — but GIA adds new edges after injecting nodes, expanding the receptive field, so that method does not carry over. Collective certification had to be rebuilt for injection attacks from the ground up.

From a worst-case attacker to a linear program

The certificate builds on node-aware bi-smoothing, a randomized-smoothing scheme that randomly deletes edges and nodes to blur the attacker's influence before classifying the smoothed graph. The key lever is locality: an injected node can only harm a target if at least one message-passing path between them survives the smoothing. The authors upper-bound the probability of that message interference, $p(E_v)$, and turn it into a concrete condition under which a target node stays certifiably robust.

To certify a whole set jointly, they then pose the adversarial question directly: under a budget of at most p injected nodes, each allowed at most τ edges, how many target nodes can the worst-case attacker break at once? That budget-allocation problem is a binary integer quadratic constrained linear program (BQCLP) — and it is NP-hard.

The paper's engineering contribution is to relax this hard integer program into an ordinary linear program (LP) that a solver handles in seconds. Because the LP has a larger feasible region than the integer program, its answer is always a sound lower bound on the true certified ratio — the certificate never over-promises. Two relaxations are offered. Collective-LP1 applies a standard linearization of the quadratic term and needs $O(p^2|T|)$ extra variables; Collective-LP2 is a customized reformulation that collapses the quadratic term first, cutting the extra variables to $O(p|T|)$ while improving both solution quality and speed.

How much does collective certification buy?

The authors evaluate on two standard citation graphs — Cora-ML (2,810 nodes, 7,981 edges, 7 classes) and Citeseer (2,110 nodes, 3,668 edges, 6 classes) — with GCN and GAT backbones (hidden size 64). The injected-node budget p is swept from 20 up to 160, with the per-node edge limit τ set to each graph's average degree (6 for Cora-ML, 4 for Citeseer). The smoothed classifier is estimated from $N = 100,000$ randomized samples (Monte Carlo) at confidence $\alpha = 0.01$, and every LP is solved with MOSEK through CVXPY.

The headline is a night-and-day gap. As the number of injected nodes grows, the sample-wise baseline collapses to zero certified nodes, while both collective certificates hold their ground.

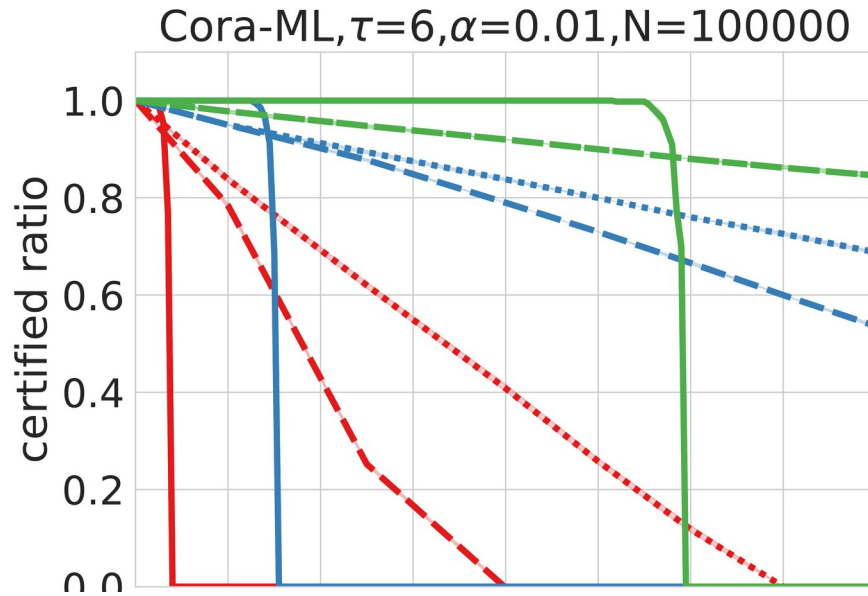


Figure 2. Certified ratio versus the injected-node budget p on Cora-ML ($\tau = 6$, $\alpha = 0.01$, $N = 100,000$). The sample-wise curves plunge to 0 at small p , while the collective curves keep certifying a large fraction of targets under heavy attack, with Collective-LP2 sitting on top.

On Citeseer with $p = 140$ injected nodes, the sample-wise certificate certifies 0.0% of targets, while Collective-LP1 certifies 73.0% and the customized Collective-LP2 reaches 81.2%. The same pattern holds on Cora-ML, shown below.

Table 1. Certified ratio on Cora-ML ($\tau = 6$, $p_e = 0.7$, $p_n = 0.9$) as the injected-node budget p grows. Sample-wise certification drops to 0.000 almost immediately, while Collective-LP2 stays well above both alternatives.

Method	$p = 20$	$p = 50$	$p = 100$	$p = 120$	$p = 140$
Sample-wise	1.000	0.000	0.000	0.000	0.000
Collective-LP1	0.920	0.768	0.452	0.316	0.178
Collective-LP2	0.926	0.836	0.686	0.624	0.564

Collective-LP2 consistently matches or beats Collective-LP1; in one Cora-ML setting ($p_e = 0.7$, $p_n = 0.9$, $p = 140$) it improves the certified ratio by 216% relative to LP1.

Fast enough to be practical

A certificate is only useful if it can actually be computed, and here the customized relaxation pays off a second time — in runtime.

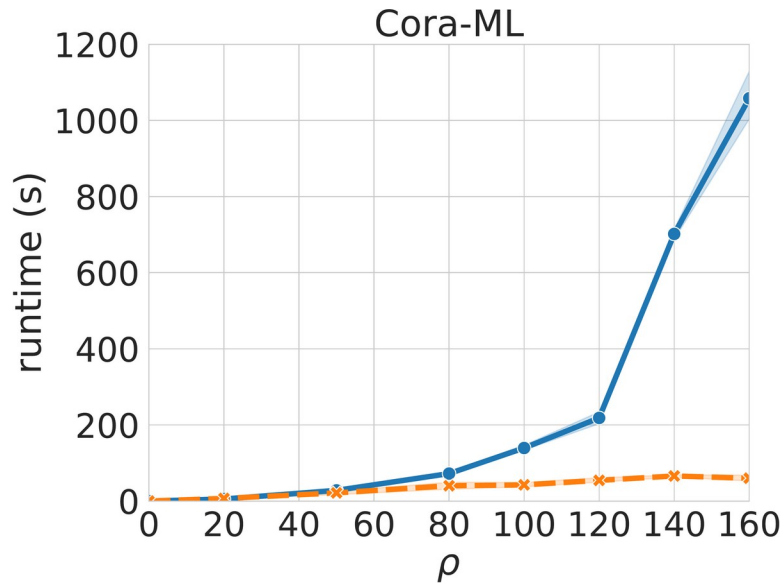


Figure 4. Runtime of the two collective LP models on Cora-ML. Collective-LP1 (upper curve) grows super-linearly with ρ , exceeding 1000 seconds at large budgets, while Collective-LP2 (lower curve) stays nearly flat near 1 minute.

Collective-LP1's runtime scales super-linearly, climbing past 1000 seconds at large budgets, whereas Collective-LP2 stays near 1 minute even at $\rho = 140$ — roughly 700 seconds faster at the high end. Solving a certificate for thousands of nodes in about 1 minute is what makes the collective scheme practical rather than merely theoretical.

What the relaxation costs

Relaxing an NP-hard integer program to an LP raises an obvious worry: how loose is the bound? To measure it, the authors compare Collective-LP2 against the exact integer BQCLP, which is only tractable for very small budgets ($\rho \leq 12$). The gap turns out to be small — about 5% in certified ratio. That same 5% loss explains the one place collective certification can trail sample-wise: at very small ρ , where sample-wise is already near-perfect and the relaxation's cost becomes visible.

Why it matters

The lesson is compact: certifying a set of nodes together, rather than one at a time, turns a near-useless 0.0% guarantee into an 80%+ certified ratio against graph injection attacks, and a customized LP relaxation keeps it solvable in about 1 minute. Because it reuses the same smoothed classifier as sample-wise certification, the collective scheme composes with existing certificates rather than replacing them.

The evaluation covers two citation graphs and message-passing GNN models, and the certificate inherits randomized smoothing's assumptions; scaling to much larger graphs and other domains is left for future work. Even so, moving from 0% to over 80% is a real step toward provable defenses that hold up under realistic, single-graph injection attacks.